



Elisa Sabadini

Date of birth: 30/01/2002

Place of birth: Carrara (MS), Italy

Nationality: Italian



europass

ABOUT ME

I would describe myself as a methodical and persistent student with a creative mindset and strong enthusiasm for research.

My main interests lie in Theoretical Physics, especially for its elegant and versatile mathematical structures, exact results, and analytical solutions.

For these reasons, I see contemporary Statistical Physics as a highly dynamic, multifaceted and exploratory field, with many new applications and problems lacking a well-established framework or approach, which is perfectly aligned with my curiosity and interests.

My background is primarily focused on quantum and classical field theories as mathematical tools, in both high-energy physics (e.g. the Standard Model, QCD) and statistical physics, with some exposure to arbitrary-spin fields in curved spacetime. In particular I am familiar with path integrals and generating functionals, the renormalization group in various contexts, effective field theories, spontaneous symmetry breaking, and aspects of anomalies.

My education also covers the traditional classical and quantum statistical physics topics beyond field-theory approaches, such as lattice models and related techniques, phase transitions and critical phenomena.

In addition, even if I am more enthusiastic about analytical computations, I have experience in using numerical techniques such as Markov chain Monte Carlo methods, exact and partial diagonalization techniques (Lanczos, Krylov space, Jacobi-Davidson), MPSs and DMRG.

Moreover, for my thesis "Quantum Quench Dynamics from Increasingly Entangled Initial States in the XXZ Model" have mainly worked on Yang-Baxter integrability and in particular on the following topics:

- Out-of-equilibrium interacting quantum many-body systems in 1D lattices (e.g. homogeneous quenches and bipartitioning protocols);
- Entanglement dynamics;
- Thermodynamic Bethe Ansatz and local relaxation;
- Quantum Inverse Scattering Method / Algebraic Bethe Ansatz (in particular quasilocal charges and string-charge duality);
- Generalized Hydrodynamics and transport;
- MPS, DMRG, tDMRG.

EDUCATION & TRAINING

13/09/2021 - 12/09/2024 Pisa, Italy

Bachelor's Degree in Physics University of Pisa

Final grade: 110/110 cum laude | **Thesis:** The second Quantization - Supervisor: Ettore Vicari

16/09/2024 - 21/07/2026 Pisa, Italy

Master's Degree in Theoretical Physics University of Pisa

Final grade: 110/110 cum laude | **Thesis:** Quantum Quench Dynamics from Increasingly Entangled Initial States in the XXZ Model - Supervisor: Vincenzo Alba

01/10/2026 - CURRENT Trieste, Italy

PhD in Statistical Physics SISSA

CONFERENCES & SEMINARS

02/02/2026 - 13/02/2026 Galileo Galilei Institute - Florence (Italy)

GGI - SFT 2026 - Lectures on Statistical Field Theories

INFN - Pisa (Italy)

INFN Theory Group Seminars

SKILLS

DIGITAL SKILLS

LaTeX | C | Fortran | Python | Mathematica

LANGUAGE SKILLS

Italian (native) | Spanish (A2) | English (C1)

HONOURS AND AWARDS

International Earth Science Olympiad (IESO), Earth Systems Art Category - 1st Place | 2021

Italian National Olympiad in Natural Sciences - 1st Place | 2021

Italian National Women's Team Mathematical Olympiad - 1st Place | 2020

Italian National Team Championship in Logic Games (5th Edition) - 1st Place | 2018

Italian National Team Final, Kangaroo Mathematics Competition - 1st Place | 2017